Home Lab 9: Build a Density Column

Do this with a parent. Goggles on. Clean up after. No heating.

Goal

Build a layered liquid column and use it to sort plastic bits by density — the same idea as the density-ladder test at competition.

You need

From home: a tall clear glass or jar. Liquids: **rubbing (isopropyl) alcohol, vegetable oil, water, and saturated salt water** (stir salt into warm water until no more dissolves, then let it cool). Optional: dish soap, honey or corn syrup for a denser layer. A few small pieces of different plastics: a bit of a **grocery bag** (LDPE, fold into a pellet), a **bottle cap** (PP), a chunk of a **clear rigid cup or CD case** (PS), a piece of a **soda bottle** (PETE). Tweezers.

Part 1: The column

Pour slowly down the side of the glass, densest first, letting each settle:

1. Saturated salt water (bottom)
2. Water
3. Vegetable oil
4. Rubbing alcohol (top) — pour very gently or it mixes with the oil

You should get visible layers. (If alcohol and oil blend, that's OK — you'll still have salt water / water / oil bands.)

Part 2: Drop in the plastics

Drop each plastic piece in with tweezers. Note which layer it settles in — that tells you its density is **between** the layer above (less dense) and the layer below (more dense).

Results

Layer densities (g/cm³): alcohol ≈ 0.79 · oil ≈ 0.92 · water ≈ 1.00 · sat. salt water ≈ 1.20

Plastic piece	Settles at / between	Density range	Likely plastic
Grocery bag			
Bottle cap			
Clear cup / CD case			
Soda bottle			

Follow-up

1. Which plastics floated on the water layer? What do those have in common?
2. The bottle-cap plastic floated even on the oil. Roughly what is its density, and which plastic is it?
3. The soda-bottle plastic sank through everything. Is its density above or below 1.20? Which plastic is it?
4. Why does foam packing "peanut" float on everything, even though solid polystyrene sinks in water?